

Kostadin Devedzhiev

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SUMMARY

PhD student at the University of Cambridge (CHIA, TRACE Lab) researching **human–AI orchestration**. Previously three years at Stellar Cyber building AI security products shown at RSA and Black Hat.

EDUCATION

- University of Cambridge** Oct 2026 – Present
PhD in Human-Inspired Artificial Intelligence · Queens' College Cambridge, UK
- **Human–AI orchestration:** multi-agent systems in which AI agents and people work together under varying expertise, cost, and availability.
 - Centre for Human-Inspired AI (CHIA), TRACE Lab; co-supervised by Prof. Umang Bhatt and Prof. Adrian Weller
 - Cambridge AI Research Society
- University of Cambridge** Oct 2025 – Jun 2026
MPhil in Human-Inspired Artificial Intelligence · Homerton College Cambridge, UK
- **Awarded with Distinction** · 82.6/100
 - Centre for Human-Inspired AI (CHIA), TRACE Lab; supervised by Prof. Umang Bhatt
 - Thesis: *Ad Hoc Human–AI Orchestration via Skill Inference*
- Stony Brook University** Aug 2018 – May 2022
BS in Computer Science & Applied Mathematics and Statistics Stony Brook, NY
- Specialisation: AI & Data Science · CS Honors Program · GPA 3.89/4.00 · Summa Cum Laude
- University of Hawai'i at Hilo** Aug 2019 – May 2020
National Student Exchange Program · Computer Science Hilo, HI
- GPA 3.97/4.00

RESEARCH EXPERIENCE

- University of Cambridge** Jul 2026 – Present
Research Assistant, Centre for Human-Inspired AI (CHIA) Cambridge, UK
- Research on **Hybrid Workforces**, a collaboration between CHIA and **Accenture Singapore**.
 - Developing skill inference, the Human API, and orchestration controls.
- Stony Brook University** Aug 2021 – May 2022
NLP Research Assistant Stony Brook, NY
- Designed **Recursive QA**, a framework for generating formal-representation annotations of natural-language specifications via guided question-answering over constituency parse trees.
 - Reduced redundant annotation options using affinity clustering on Levenshtein distance; achieved **>80% inter-annotator agreement** with experienced users completing annotations in ~30 s.
 - Built a full-stack annotation tool with account management, work-history tracking, and graph visualisations.
- University of Hawai'i at Hilo** Jun 2020 – Aug 2020
AI Research Assistant Hilo, HI
- Developed an IoU/F1-based threshold estimation heuristic to optimise real-time CNN proposal acceptance for **invasive-species detection** in drone imagery.
 - Enhanced a human-in-the-loop semi-automatic annotation tool that progressively increases AI assistance as model accuracy improves during training.

PROFESSIONAL EXPERIENCE

- Stellar Cyber** May 2022 – Sep 2025
Software Engineer San Jose, CA
- Architected a **multi-agent autonomous SOC system** that triages security cases by correlating alerts with LLMs and generating analyst-ready summaries; featured as Stellar's flagship demo at **RSA 2025** and **Black Hat 2025**.
 - Built the chat interface, session management, and multi-chart visualisation suite for **Open XDR Investigator**, a GenAI copilot translating natural-language prompts into Elasticsearch queries; headlined **RSA 2024** and **Black Hat 2024**.
 - Implemented bidirectional WebSocket communication with parallel data and LLM requests, **cutting average response times by 70%** and enabling real-time progress streaming.
 - Developed knowledge-enriched GPT agents for log analysis, product metrics, connector normalisation, and data-source classification, each capable of visualisations, API calls, and script execution.

- Owned the product analytics framework (Mixpanel), producing reports that informed UX redesigns, storage optimisation, and behavioural segmentation.
- Maintained >90% test coverage on owned components, **reducing bug filings by 34%** year-over-year.

University of Hawai'i at Hilo

Mar 2020 – May 2020

Creative Electronic Media Assistant

Hilo, HI

- Built a hologram display API and developed the Data Visualization Lab website.

Vivansa

Jun 2019 – Aug 2019

Software Engineering Intern

Sofia, Bulgaria

- Improved CRM front-end components and resolved erroneous database entries through root-cause analysis and automated data-cleaning procedures.

RESEARCH & PROJECTS

Ad Hoc Human–AI Orchestration via Skill Inference – MPhil Thesis

- Models each teammate's ability as a **Bayesian belief over a taxonomy of skills**. Because the skills are correlated, measuring one also updates the others, so profiles converge on fewer observations than scoring each skill separately.
- Released as **skillinfer**, an open-source Python package on [PyPI](#) with source on [GitHub](#), and previewed in Tailor.

Tailor – Human–AI Orchestration Platform

- Visual workflow builder that combines AI agents with human review steps and four levels of governance control, built for regulated industries.

PUBLICATION

P-M Binder, **K. G. Devedzhiev**, A. T. Runyan. *Motional emf generated by squeezing an elliptical conducting loop*. 2020 European Journal of Physics. [doi:10.1088/1361-6404/abb066](#)

TALKS & PRESENTATIONS

Oxford Human–Algorithm Interaction (HAI) Workshop 2026, Saïd Business School, Oxford – Talk: *Ad Hoc Human–AI Orchestration via Skill Inference*. Jul 2026

CHIA Annual Conference 2026: AI for Good, Cambridge Union – Poster: *Ad Hoc Human–AI Orchestration via Skill Inference*. Jun 2026

ICO Technical Workshop: Know Your Data, Information Commissioner's Office, London – Participant; AI transparency and personal data in foundation-model training. Mar 2026

TEACHING & MENTORSHIP

AI Leadership Mentor, [UniHawk](#), Kuwait City – Mentored high-school students through a week-long AI app-building bootcamp. Dec 2025

Teaching Assistant, Stony Brook University – AMS 210: Applied Linear Algebra; held weekly office hours and graded coursework. Aug 2020 – May 2021

Linear Algebra Grader, University of Hawai'i at Hilo – MATH 311: Linear Algebra.

Mar 2020 – May 2020

Computer Science Grader, University of Hawai'i at Hilo – CS 150: Intro to Computer Science.

Oct 2019 – Dec 2019

CERTIFICATIONS

IBM Generative AI Engineering – IBM

SKILLS

Domains	Machine Learning, NLP, Human-Computer Interaction, Data Science, Full-Stack Development
Languages	Python, JavaScript, TypeScript, C
ML & AI	PyTorch, Hugging Face, Scikit-learn, LangChain, LangGraph, Pandas, NumPy, Plotly
Web	React, Angular, Tailwind, FastAPI, Node.js, Express, Flask
Data & Infra	PostgreSQL, MongoDB, Elasticsearch, Pinecone, Redis, Docker